

Initiating Search

November 4, 2025, 4:58 PM

• Search:

CAS Lexicon Search:

Ullmann reaction - Preferred Concept

Search Tasks

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Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	
Publication Year:	2020 to 2025	

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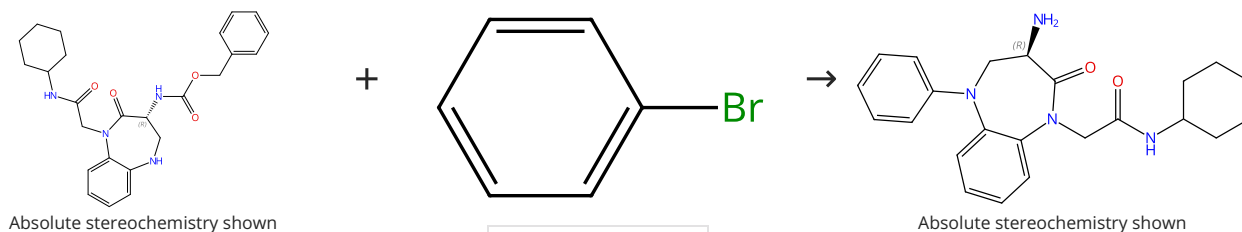
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (81)

31-614-CAS-24424602

Steps: 1 Yield: 100%

Efficient synthesis of polysubstituted 1,5-benzodiazepinone dipeptide mimetics via an Ugi-4CR-Ullmann condensation sequence

1.1 Reagents: Hydrochloric acid
Solvents: 1,4-Dioxane; 1 h, rt

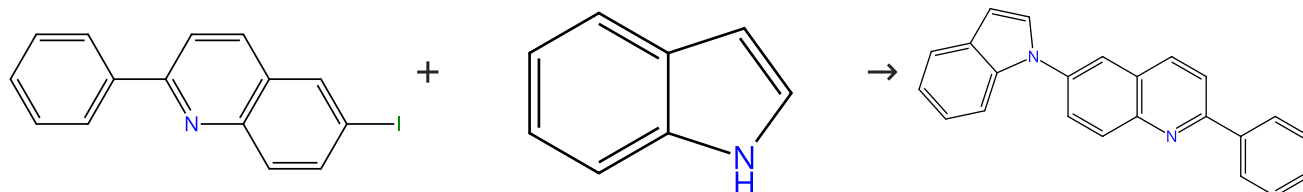
Experimental Protocols

By: Van Den Hauwe, Robin; et al

Synlett (2021), 32(17), 1719-1724.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

Suppliers (114)

31-172-CAS-23026515

Steps: 1 Yield: 99%

The remarkable selectivity of the 2-arylquinoline-based acyl hydrazones toward copper salts: exploration of their catalytic applications in the copper catalyzed N-arylation of indole derivatives and C1-alkynylation of tetrahydroisoquinolines via the A³ reaction

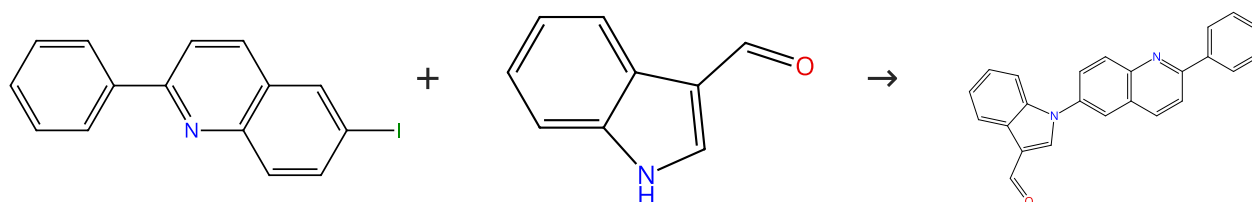
1.1 Reagents: Potassium carbonate
Catalysts: Cuprous iodide, 2568377-50-2
Solvents: Dimethyl sulfoxide; 24 h, 110 °C

By: Echeverry-Gonzalez, Carlos A.; et al

New Journal of Chemistry (2021), 45(1), 243-250.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 99%



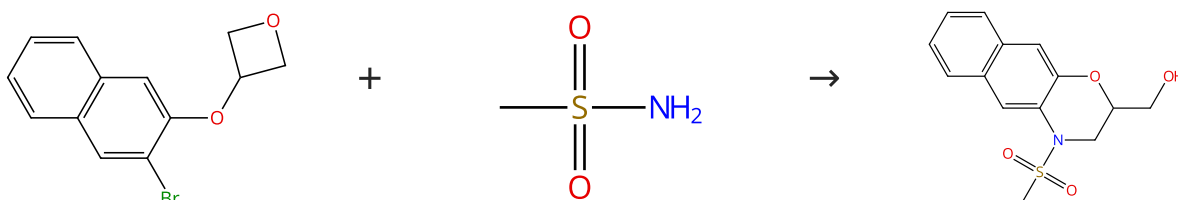
Suppliers (2)

Suppliers (116)

31-172-CAS-23028213 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Cuprous iodide, 2568377-50-2 Solvents: Dimethyl sulfoxide; 48 h, 110 °C	The remarkable selectivity of the 2-arylquinoline-based acyl hydrazones toward copper salts: exploration of their catalytic applications in the copper catalyzed N- arylation of indole derivatives and C1-alkynylation of tetrahydroisoquinolines via the A³ reaction By: Echeverry-Gonzalez, Carlos A.; et al New Journal of Chemistry (2021), 45(1), 243-250.
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Scheme 4 (1 Reaction)

Steps: 1 Yield: 99%

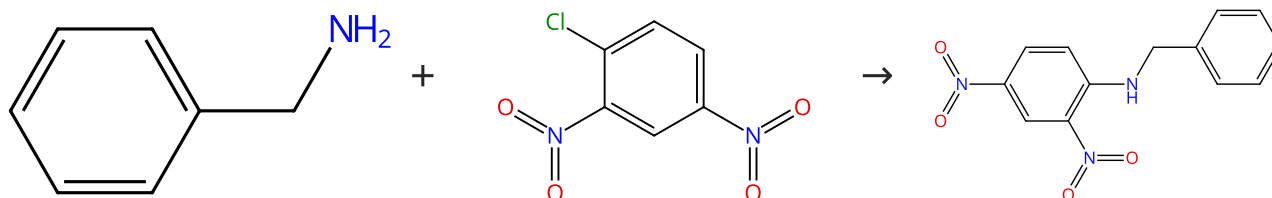


Suppliers (80)

31-614-CAS-24755323 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][dicyclohexyl [3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine-κP](methanesulfonato-κO)palladium Solvents: 1,4-Dioxane; 16 h, 120 °C Experimental Protocols	Tandem Amination/Oxetane Ring Opening toward Benzomorpholines By: DeRatt, Lindsey G.; et al Journal of Organic Chemistry (2021), 86(23), 17482-17486.
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Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (81)

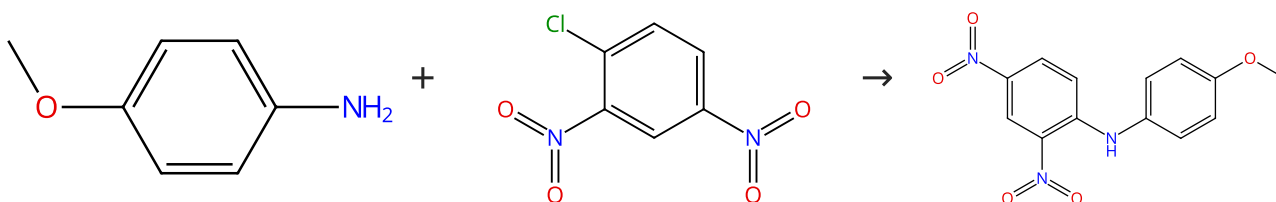
Suppliers (76)

Suppliers (30)

31-614-CAS-34769822 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Copper Solvents: Tetrahydrofuran; 60 min, 65 °C Experimental Protocols	Magnetic MCR-Functionalized Graphene Oxide Complexed with Copper Nano-Particles: An Efficient and Recyclable Nanocatalyst for Ullmann C-N Coupling Reaction By: Bagheri, Hashem; et al Polycyclic Aromatic Compounds (2023), 43(8), 7580-7596.
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Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

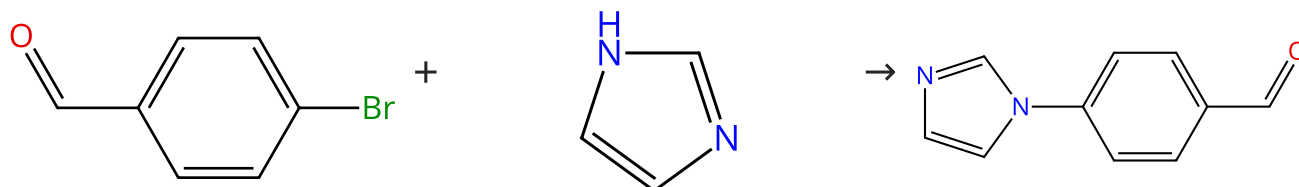
Suppliers (76)

Suppliers (16)

31-614-CAS-34769830	Steps: 1 Yield: 99%	Magnetic MCR-Functionalized Graphene Oxide Complexed with Copper Nano-Particles: An Efficient and Recyclable Nanocatalyst for Ullmann C-N Coupling Reaction
1.1 Reagents: Potassium carbonate Catalysts: Copper Solvents: Tetrahydrofuran; 60 min, 65 °C		By: Bagheri, Hashem; et al
Experimental Protocols		Polycyclic Aromatic Compounds (2023), 43(8), 7580-7596.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (91)

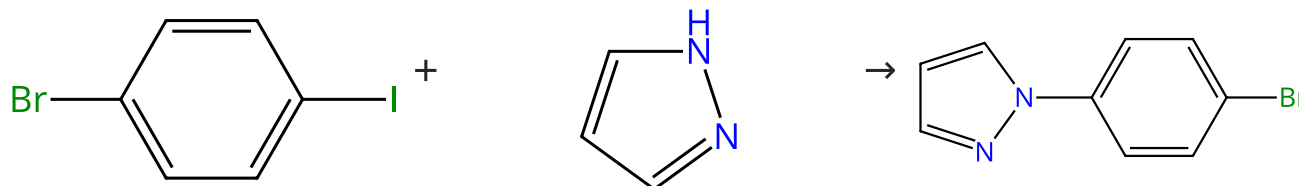
Suppliers (206)

Suppliers (89)

31-614-CAS-38094317	Steps: 1 Yield: 99%	Nucleation of silver nanoparticles by tetrafacial organic molecular barrel: Potential use in visible-light-triggered photocatalysis
1.1 Reagents: Potassium carbonate Catalysts: Silver, Benzaldehyde, 4,4',4'',4'''-(1,2,4,5-benzene tetrayl)tetrakis-, Schiff base with 1,2-ethanediamine (4:8), dotriacontahydro deriv. Solvents: Dimethylformamide; 10 h, 30 °C		By: Mondal, Bijnaneswar; et al
Experimental Protocols		Chemistry - A European Journal (2020), 26(65), 15007-15015.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (92)

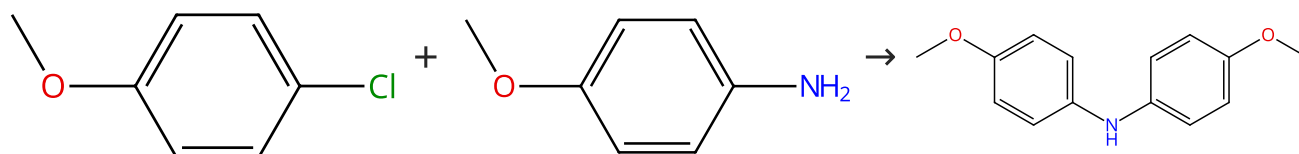
Suppliers (94)

Suppliers (60)

31-172-CAS-22770666	Steps: 1 Yield: 99%	Activation of Copper Species on Carbon Nitride for Enhanced Activity in the Arylation of Amines
1.1 Reagents: Cesium carbonate Catalysts: Iron, Cuprous iodide, Carbon nitride (C ₃ N ₄) Solvents: Dimethylformamide; 12 h, rt → 393 K		By: Vorobyeva, Evgeniya; et al
		ACS Catalysis (2020), 10(19), 11069-11080.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (72)

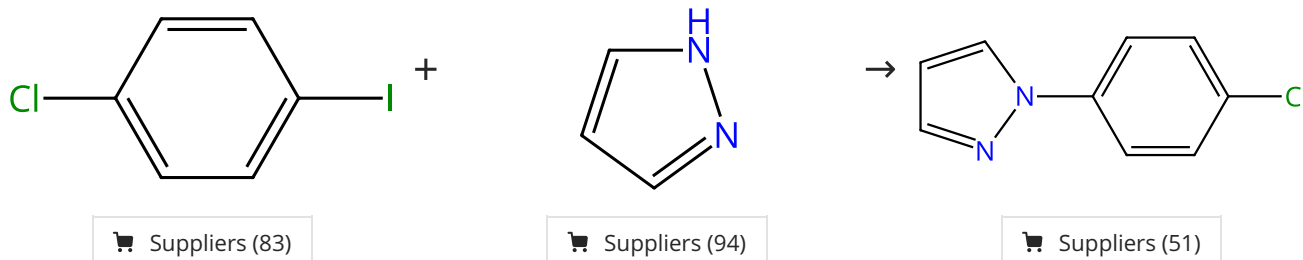
Suppliers (83)

Suppliers (69)

31-614-CAS-35773757	Steps: 1 Yield: 99%	Design and application of a novel and effective ligand for the Cu-catalyzed amination of aryl halides in water By: Wang, Lifang; et al Green Chemistry Letters and Reviews (2023), 16(1), 2147027/1-2147027/7.
1.1 Reagents: Potassium hydroxide Catalysts: Copper oxide (CuO), Tetrabutylammonium bromide, 1 <i>H</i> -Pyrrole-2-carboxylic acid, 2,2-bis[2-(hydroxymethyl)phenyl]hydrazide Solvents: Water; 12 h, 130 °C		
Experimental Protocols		

Scheme 10 (1 Reaction)

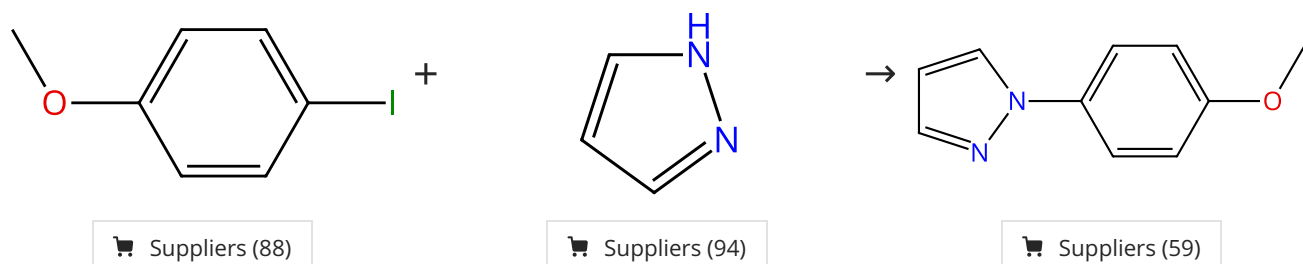
Steps: 1 Yield: 99%



31-172-CAS-22770665	Steps: 1 Yield: 99%	Activation of Copper Species on Carbon Nitride for Enhanced Activity in the Arylation of Amines By: Vorobyeva, Evgeniya; et al ACS Catalysis (2020), 10(19), 11069-11080.
1.1 Reagents: Cesium carbonate Catalysts: Iron, Cuprous iodide, Carbon nitride (C ₃ N ₄) Solvents: Dimethylformamide; 12 h, rt → 393 K		

Scheme 11 (1 Reaction)

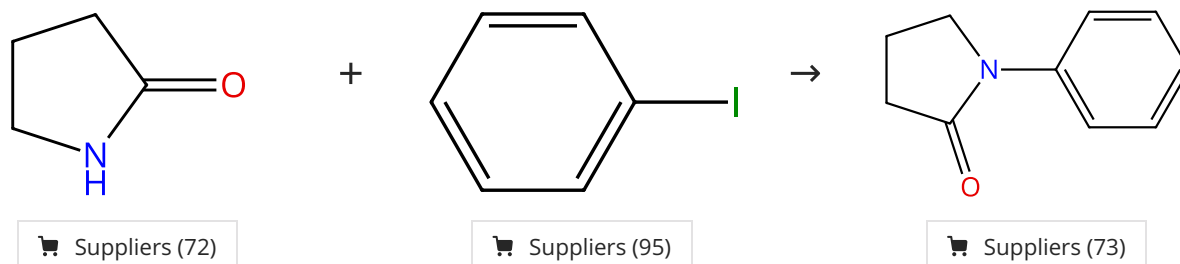
Steps: 1 Yield: 99%



31-172-CAS-22770662	Steps: 1 Yield: 99%	Activation of Copper Species on Carbon Nitride for Enhanced Activity in the Arylation of Amines By: Vorobyeva, Evgeniya; et al ACS Catalysis (2020), 10(19), 11069-11080.
1.1 Reagents: Cesium carbonate Catalysts: Iron, Cuprous iodide, Carbon nitride (C ₃ N ₄) Solvents: Dimethylformamide; 12 h, rt → 393 K		

Scheme 12 (1 Reaction)

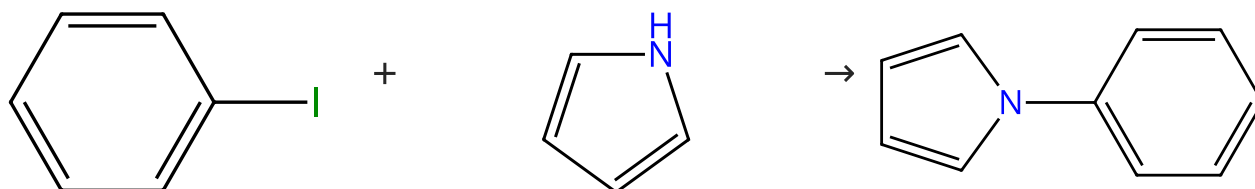
Steps: 1 Yield: 99%



31-172-CAS-22770673	Steps: 1 Yield: 99%	Activation of Copper Species on Carbon Nitride for Enhanced Activity in the Arylation of Amines
1.1 Reagents: Cesium carbonate Catalysts: Iron, Cuprous iodide, Carbon nitride (C ₃ N ₄) Solvents: Dimethylformamide; 12 h, rt → 393 K		By: Vorobyeva, Evgeniya; et al ACS Catalysis (2020), 10(19), 11069-11080.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

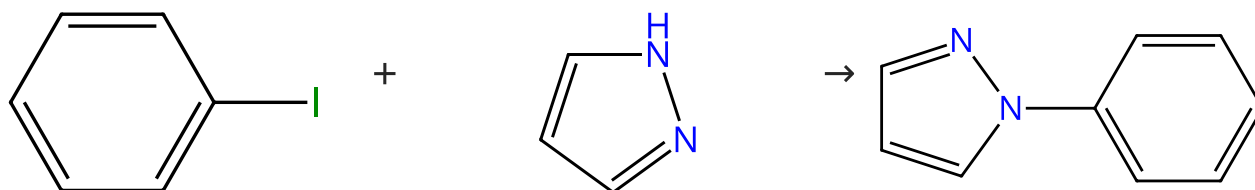
Suppliers (76)

Suppliers (71)

31-172-CAS-23298238	Steps: 1 Yield: 99%	L-Proline N-oxide dihydrazides as an efficient ligand for cross-coupling reactions of aryl iodides and bromides with amines and phenols
1.1 Reagents: Cesium carbonate Catalysts: Cuprous iodide, 2588211-74-7 Solvents: Ethanol; 12 h, 80 °C		By: Ding, Zhiqiang; et al Tetrahedron (2021), 79, 131826.
Experimental Protocols		

Scheme 14 (2 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (95)

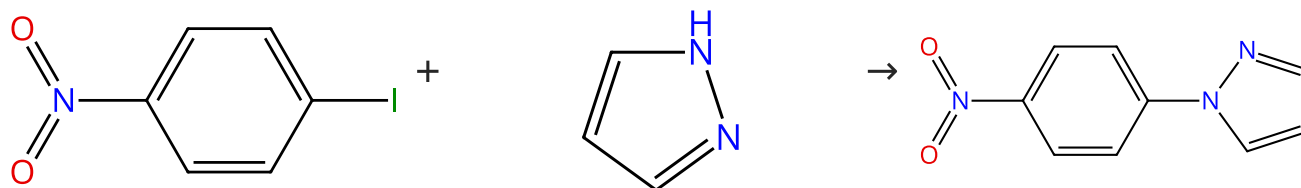
Suppliers (94)

Suppliers (85)

31-172-CAS-22770660	Steps: 1 Yield: 99%	Activation of Copper Species on Carbon Nitride for Enhanced Activity in the Arylation of Amines
1.1 Reagents: Cesium carbonate Catalysts: Iron, Cuprous iodide, Carbon nitride (C ₃ N ₄) Solvents: Dimethylformamide; 12 h, rt → 393 K		By: Vorobyeva, Evgeniya; et al ACS Catalysis (2020), 10(19), 11069-11080.
31-172-CAS-23297637	Steps: 1 Yield: 98%	L-Proline N-oxide dihydrazides as an efficient ligand for cross-coupling reactions of aryl iodides and bromides with amines and phenols
1.1 Reagents: Cesium carbonate Catalysts: Cuprous iodide, 2588211-74-7 Solvents: Ethanol; 12 h, 80 °C		By: Ding, Zhiqiang; et al Tetrahedron (2021), 79, 131826.
Experimental Protocols		

Scheme 15 (2 Reactions)

Steps: 1 Yield: 97-99%



Suppliers (70)

Suppliers (94)

Suppliers (57)

31-172-CAS-22770664

Steps: 1 Yield: 99%

Activation of Copper Species on Carbon Nitride for Enhanced Activity in the Arylation of Amines

1.1 **Reagents:** Cesium carbonate
Catalysts: Iron, Cuprous iodide, Carbon nitride (C₃N₄)
Solvents: Dimethylformamide; 12 h, rt → 393 K

By: Vorobyeva, Evgeniya; et al

ACS Catalysis (2020), 10(19), 11069-11080.

31-172-CAS-21455541

Steps: 1 Yield: 97%

Preparation, Characterization and Application of Copper Schiff base Complex Supported on MCM-41 as a Recyclable Catalyst for the Ullmann-type N-arylation Reaction

1.1 **Reagents:** Potassium carbonate
Catalysts: 2409651-49-4
Solvents: Dimethylformamide, Water; 3 h, 85 °C

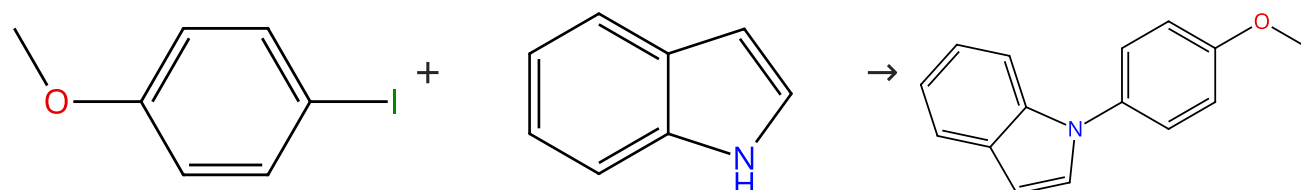
By: Niakan, Mahsa; et al

ChemistrySelect (2020), 5(1), 40-48.

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (88)

Suppliers (114)

Suppliers (11)

31-172-CAS-23033834

Steps: 1 Yield: 99%

The remarkable selectivity of the 2-arylquinoline-based acyl hydrazones toward copper salts: exploration of their catalytic applications in the copper catalyzed N-arylation of indole derivatives and C1-alkynylation of tetrahydroisoquinolines via the A³ reaction

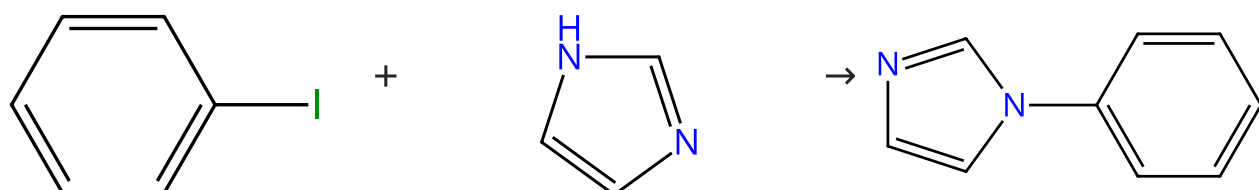
1.1 **Reagents:** Potassium carbonate
Catalysts: Cuprous iodide, 2568377-50-2
Solvents: Dimethyl sulfoxide; 24 h, 110 °C

By: Echeverry-Gonzalez, Carlos A.; et al

New Journal of Chemistry (2021), 45(1), 243-250.

Scheme 17 (2 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (95)

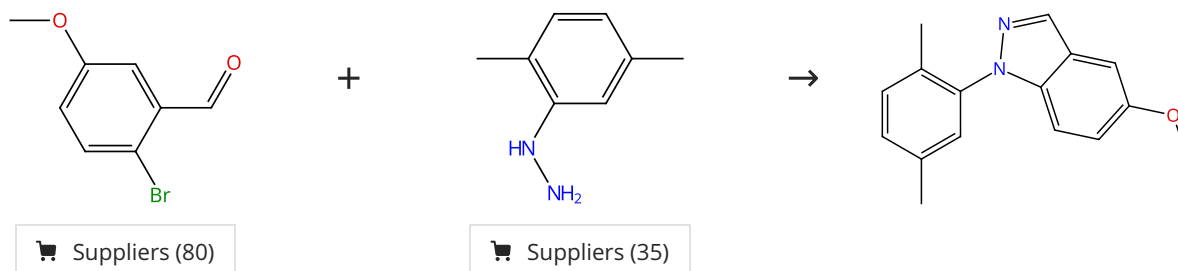
Suppliers (206)

Suppliers (80)

31-172-CAS-23295786 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Catalysts: Cuprous iodide, 2588211-74-7 Solvents: Ethanol; 12 h, 80 °C Experimental Protocols	L-Proline N-oxide dihydrazides as an efficient ligand for cross-coupling reactions of aryl iodides and bromides with amines and phenols By: Ding, Zhiqiang; et al Tetrahedron (2021), 79, 131826.
31-172-CAS-21750791 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Catalysts: Copper Solvents: Water; rt → 100 °C; 24 h, 100 °C Experimental Protocols	Facile synthesis of hydrochar supported copper nanocatalyst for Ullmann C-N coupling reaction in water By: Ge, Xin; et al Molecular Catalysis (2020), 484, 110726.

Scheme 18 (1 Reaction)

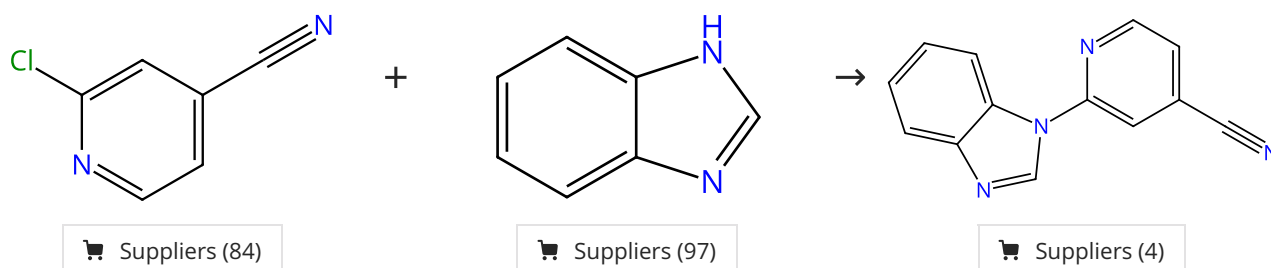
Steps: 1 Yield: 98%



31-614-CAS-34659799 Steps: 1 Yield: 98% 1.1 Catalysts: 2832920-74-6 (support palladium and copper) Solvents: Dimethylformamide; 10 min, rt 1.2 Reagents: Cesium carbonate; 2 h, 120 °C Experimental Protocols	Bottom-Up De Novo Synthesis of Porous Organic Polymers with Enone Functionalities as Supports for Pd and Cu Nanoparticles for Catalytic Tandem Synthesis By: Yadav, Chetna; et al ACS Applied Nano Materials (2022), 5(10), 14296-14310.
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Scheme 19 (1 Reaction)

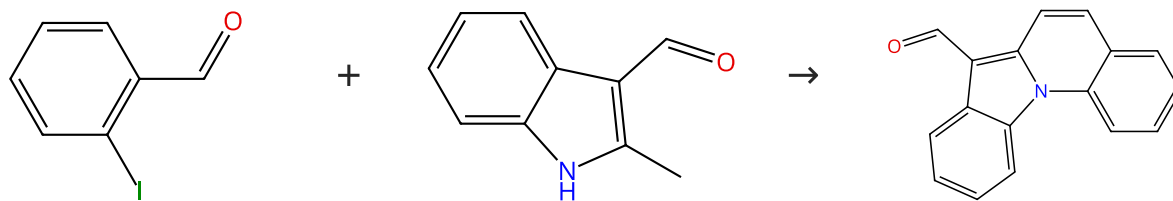
Steps: 1 Yield: 98%



31-614-CAS-40820777 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Catalysts: 3048070-21-6 Solvents: Dimethylformamide; 1.5 h, 80 °C Experimental Protocols	A new highly efficient NHC/Cu(I) catalyst system for the Ullmann-type N-arylation reactions By: Liu, Guiyan; et al Molecular Catalysis (2024), 564, 114284.
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Scheme 20 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (83)

Suppliers (83)

31-614-CAS-24155185

Steps: 1 Yield: 98%

Copper-catalyzed synthesis and anticancer activity evaluation of indolo[1,2-a]quinoline derivatives

By: Thanetchaiyakup, Adisak; et al

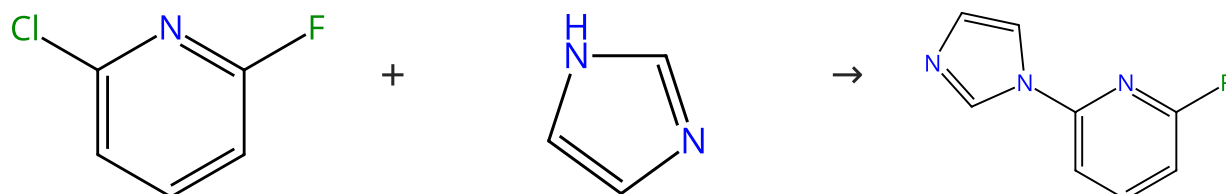
Tetrahedron Letters (2021), 82, 153365.

1.1 **Reagents:** Cesium carbonate
Catalysts: L-Proline, Cuprous iodide
Solvents: Dimethyl sulfoxide; 6 h, 100 °C

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (70)

Suppliers (206)

Suppliers (31)

31-614-CAS-40820774

Steps: 1 Yield: 98%

A new highly efficient NHC/Cu(I) catalyst system for the Ullmann-type N-arylation reactions

By: Liu, Guiyan; et al

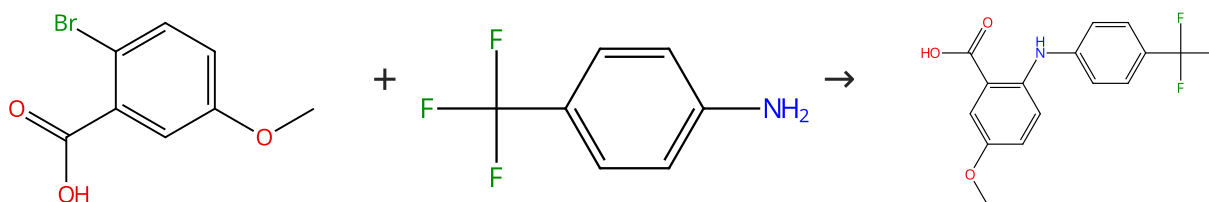
Molecular Catalysis (2024), 564, 114284.

1.1 **Reagents:** Cesium carbonate
Catalysts: Cuprous iodide, Imidazo[1,5-a][1,10]phenanthroline, 10-[2,6-bis(1-methylethyl)phenyl]-, chloride (1:1)
Solvents: Dimethylformamide; 3 h, 80 °C

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (96)

Suppliers (86)

31-614-CAS-25102541

Steps: 1 Yield: 98%

Thermochemiluminescence-Based Sensitive Probes: Synthesis and Photophysical Characterization of Acridine-Containing 1,2-Dioxetanes Focusing on Fluorophore Push-Pull Effects

By: Moroni, Giada; et al

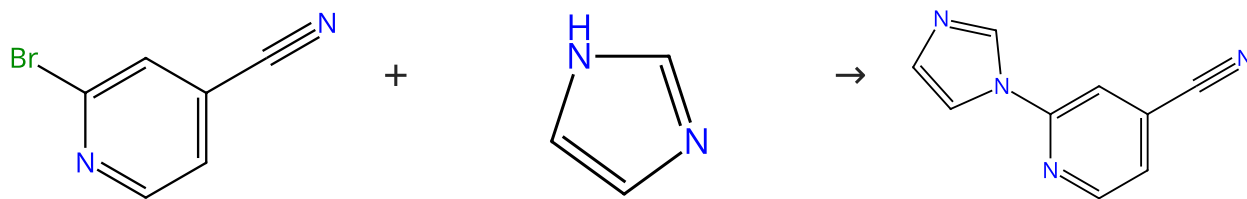
ChemPhotoChem (2022), 6(1), e202100152.

1.1 **Catalysts:** Tris(dibenzylideneacetone)dipalladium, BINAP
Solvents: Toluene; 30 min, 25 °C
 1.2 **Reagents:** Cesium carbonate; 110 - 130 °C; 130 °C → 25 °C
 1.3 **Reagents:** Sodium hydroxide
Solvents: Methanol, Water; 25 °C
 1.4 **Reagents:** Hydrochloric acid
Solvents: Water; pH 3, 25 °C

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (81)

Suppliers (206)

Suppliers (46)

31-614-CAS-40820781

Steps: 1 Yield: 98%

A new highly efficient NHC/Cu(I) catalyst system for the Ullmann-type N-arylation reactions

By: Liu, Guiyan; et al

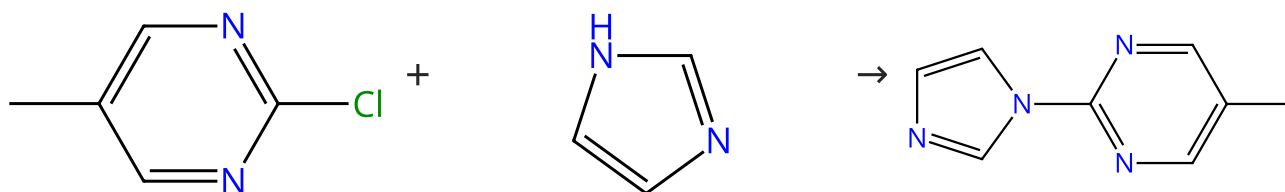
Molecular Catalysis (2024), 564, 114284.

1.1 **Reagents:** Cesium carbonate
Catalysts: Cuprous iodide, Imidazo[1,5-*a*][1,10]phenanthroline, 10-[2,6-bis(1-methylethyl)phenyl]-, chloride (1:1)
Solvents: Dimethylformamide; 3 h, 80 °C

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (85)

Suppliers (206)

Suppliers (5)

31-614-CAS-40820778

Steps: 1 Yield: 98%

A new highly efficient NHC/Cu(I) catalyst system for the Ullmann-type N-arylation reactions

By: Liu, Guiyan; et al

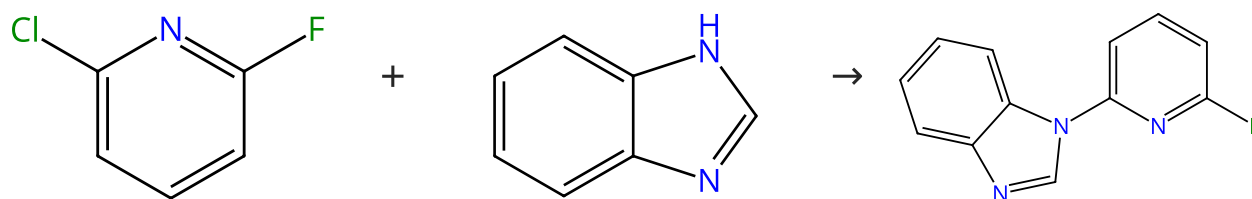
Molecular Catalysis (2024), 564, 114284.

1.1 **Reagents:** Cesium carbonate
Catalysts: 3048070-21-6
Solvents: Dimethylformamide; 1.5 h, 80 °C

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (70)

Suppliers (97)

Suppliers (4)

31-614-CAS-40820771

Steps: 1 Yield: 98%

A new highly efficient NHC/Cu(I) catalyst system for the Ullmann-type N-arylation reactions

By: Liu, Guiyan; et al

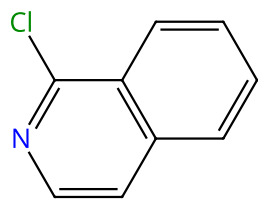
Molecular Catalysis (2024), 564, 114284.

1.1 **Reagents:** Cesium carbonate
Catalysts: 3048070-21-6
Solvents: Dimethylformamide; 1.5 h, 80 °C

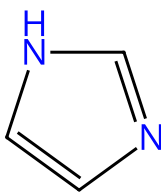
Experimental Protocols

Scheme 26 (1 Reaction)

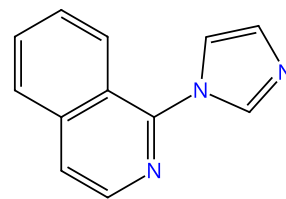
Steps: 1 Yield: 98%



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Suppliers (94)

Suppliers (206)

Suppliers (8)

31-614-CAS-40820768

Steps: 1 Yield: 98%

A new highly efficient NHC/Cu(I) catalyst system for the Ullmann-type N-arylation reactions

1.1 Reagents: Cesium carbonate

Catalysts: 3048070-21-6

Solvents: Dimethylformamide; 1.5 h, 80 °C

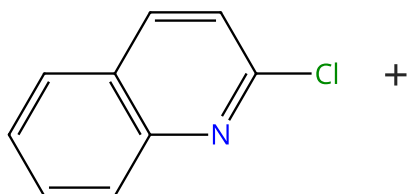
By: Liu, Guiyan; et al

Molecular Catalysis (2024), 564, 114284.

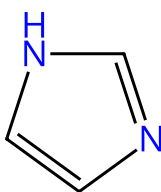
Experimental Protocols

Scheme 27 (2 Reactions)

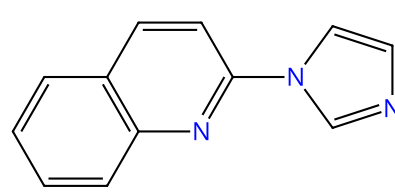
Steps: 1 Yield: 98%



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Suppliers (89)

Suppliers (206)

Suppliers (5)

31-614-CAS-40820757

Steps: 1 Yield: 98%

A new highly efficient NHC/Cu(I) catalyst system for the Ullmann-type N-arylation reactions

1.1 Reagents: Cesium carbonate

Catalysts: Cuprous iodide, Imidazo[1,5-*a*][1,10]phenanthroline, 10-[2,6-bis(1-methylethyl)phenyl]-, chloride (1:1)

Solvents: Dimethylformamide; 3 h, 80 °C

By: Liu, Guiyan; et al

Molecular Catalysis (2024), 564, 114284.

Experimental Protocols

31-172-CAS-22720808

Steps: 1 Yield: 98%

N-Heterocyclic Carbene Copper(I) Complex Catalyzed Coupling of (Hetero)aryl Chlorides and Nitrogen Heterocycles: Highly Efficient Catalytic System

1.1 Reagents: Cesium carbonate

Catalysts: [8-[2,6-Bis(1-methylethyl)phenyl]imidazo[1,5-*a*][1,8]naphthyridin-9-ylidene- κ^C, κ^N]chlorocopper

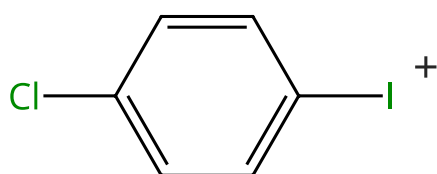
Solvents: Dimethylformamide; 12 h, 120 °C

By: Zhang, Mengyao; et al

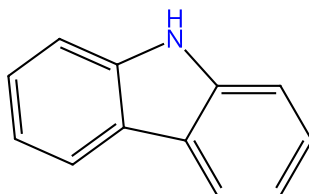
Chinese Journal of Chemistry (2020), 38(11), 1252-1256.

Scheme 28 (1 Reaction)

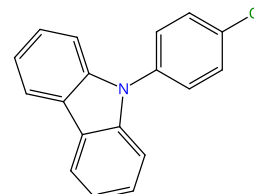
Steps: 1 Yield: 98%



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Suppliers (83)

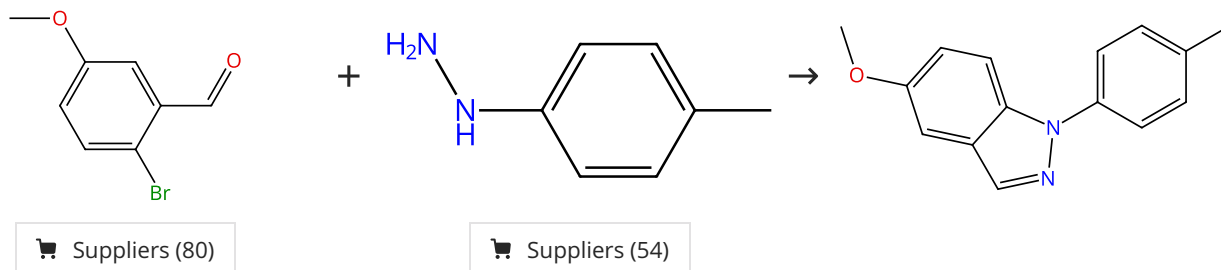
Suppliers (117)

Suppliers (10)

31-172-CAS-21455550	Steps: 1 Yield: 98%	Preparation, Characterization and Application of Copper Schiff base Complex Supported on MCM-41 as a Recyclable Catalyst for the Ullmann-type N-arylation Reaction By: Niakan, Mahsa; et al ChemistrySelect (2020), 5(1), 40-48.
1.1 Reagents: Potassium carbonate Catalysts: 2409651-49-4 Solvents: Dimethylformamide, Water; 5 h, 85 °C		
Experimental Protocols		

Scheme 29 (1 Reaction)

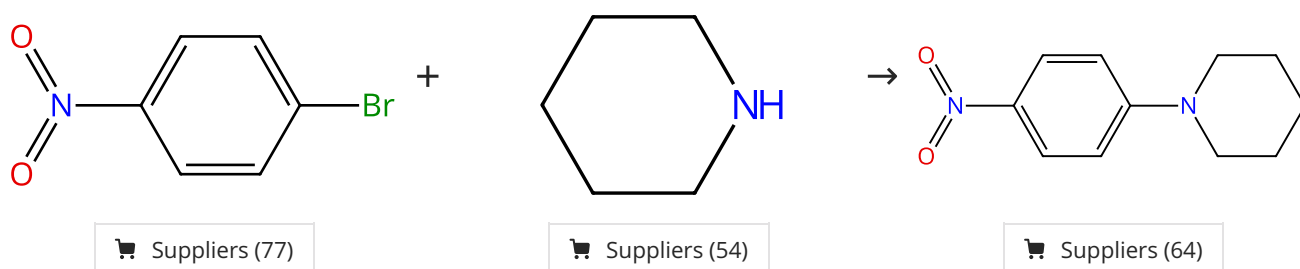
Steps: 1 Yield: 98%



31-614-CAS-34659794	Steps: 1 Yield: 98%	Bottom-Up De Novo Synthesis of Porous Organic Polymers with Enone Functionalities as Supports for Pd and Cu Nanoparticles for Catalytic Tandem Synthesis By: Yadav, Chetna; et al ACS Applied Nano Materials (2022), 5(10), 14296-14310.
1.1 Catalysts: 2832920-74-6 (support palladium and copper) Solvents: Dimethylformamide; 10 min, rt		
1.2 Reagents: Cesium carbonate; 2 h, 120 °C		
Experimental Protocols		

Scheme 30 (1 Reaction)

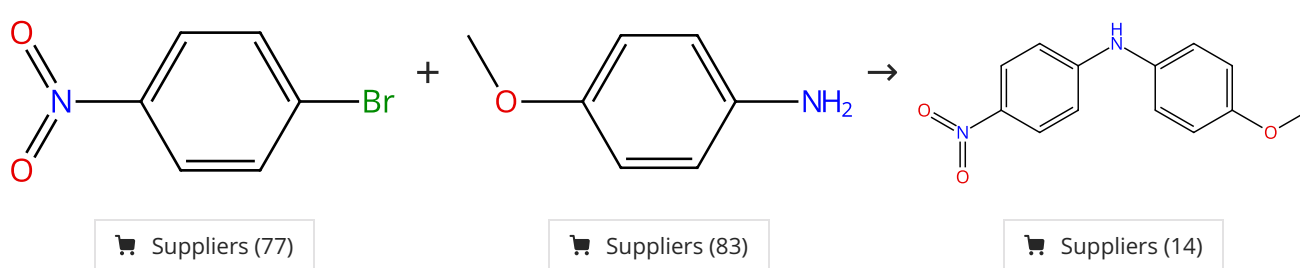
Steps: 1 Yield: 98%



31-614-CAS-34769818	Steps: 1 Yield: 98%	Magnetic MCR-Functionalized Graphene Oxide Complexed with Copper Nano-Particles: An Efficient and Recyclable Nanocatalyst for Ullmann C-N Coupling Reaction By: Bagheri, Hashem; et al Polycyclic Aromatic Compounds (2023), 43(8), 7580-7596.
1.1 Reagents: Potassium carbonate Catalysts: Copper Solvents: Tetrahydrofuran; 60 min, 65 °C		
Experimental Protocols		

Scheme 31 (1 Reaction)

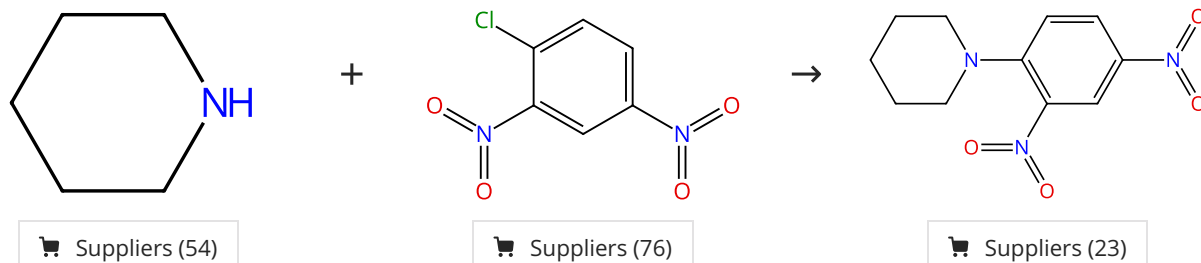
Steps: 1 Yield: 98%



31-614-CAS-34769829	Steps: 1 Yield: 98%	Magnetic MCR-Functionalized Graphene Oxide Complexed with Copper Nano-Particles: An Efficient and Recyclable Nanocatalyst for Ullmann C-N Coupling Reaction
1.1 Reagents: Potassium carbonate Catalysts: Copper Solvents: Tetrahydrofuran; 60 min, 65 °C		By: Bagheri, Hashem; et al
Experimental Protocols		Polycyclic Aromatic Compounds (2023), 43(8), 7580-7596.

Scheme 32 (1 Reaction)

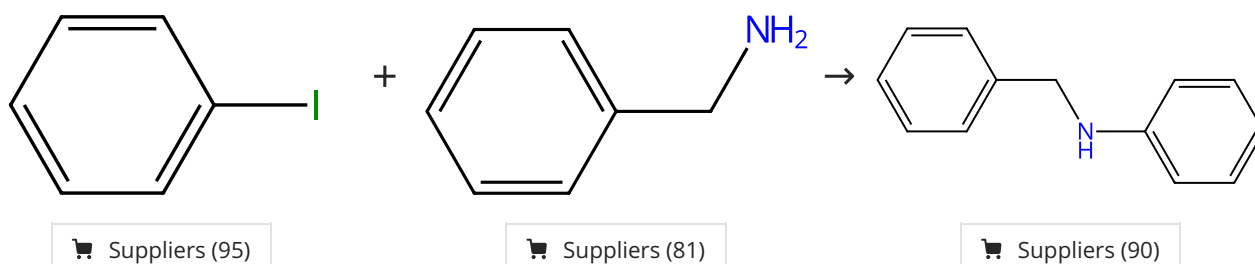
Steps: 1 Yield: 98%



31-614-CAS-34769820	Steps: 1 Yield: 98%	Magnetic MCR-Functionalized Graphene Oxide Complexed with Copper Nano-Particles: An Efficient and Recyclable Nanocatalyst for Ullmann C-N Coupling Reaction
1.1 Reagents: Potassium carbonate Catalysts: Copper Solvents: Tetrahydrofuran; 60 min, 65 °C		By: Bagheri, Hashem; et al
Experimental Protocols		Polycyclic Aromatic Compounds (2023), 43(8), 7580-7596.

Scheme 33 (3 Reactions)

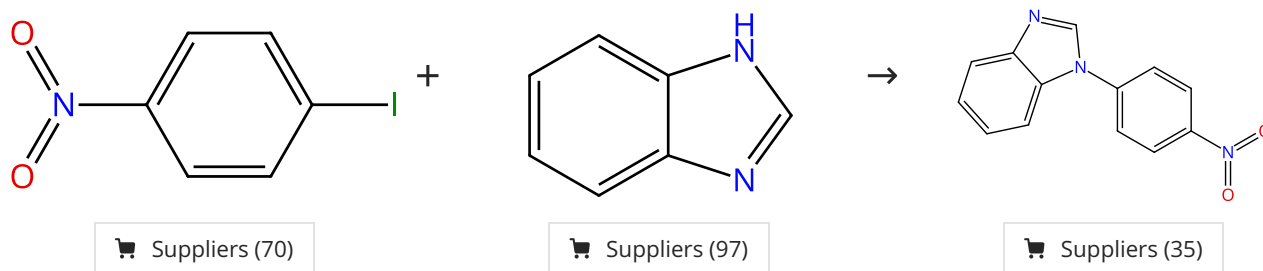
Steps: 1 Yield: 97-98%



31-614-CAS-32326813	Steps: 1 Yield: 98%	Tartramide Ligands for Copper-Catalyzed N-Arylation at Room Temperature
1.1 Reagents: Sodium hydroxide Catalysts: Cuprous iodide, Butanediamide, 2,3-dihydroxy- <i>N</i> , <i>N'</i> -bis[(4-methoxyphenyl)methyl]-, [<i>R</i> -(<i>R</i> *, <i>R</i> *)]- Solvents: Diethylene glycol; 24 h, rt		By: Ma, Xuerui; et al
Experimental Protocols		Advanced Synthesis & Catalysis (2022), 364(12), 2023-2031.
31-614-CAS-32326806	Steps: 1 Yield: 97%	Tartramide Ligands for Copper-Catalyzed N-Arylation at Room Temperature
1.1 Reagents: Sodium hydroxide Catalysts: Cuprous iodide, (2 <i>R</i> ,3 <i>R</i>)-2,3-Dihydroxy- <i>N</i> ¹ , <i>N</i> ⁴ -bis(phenylmethyl)butanediamide Solvents: Diethylene glycol; 24 h, rt		By: Ma, Xuerui; et al
Experimental Protocols		Advanced Synthesis & Catalysis (2022), 364(12), 2023-2031.
31-614-CAS-32326803	Steps: 1 Yield: 97%	Tartramide Ligands for Copper-Catalyzed N-Arylation at Room Temperature
1.1 Reagents: Sodium hydroxide Catalysts: Cuprous iodide, (2 <i>R</i> ,3 <i>R</i>)-2,3-Dihydroxy- <i>N</i> ¹ , <i>N</i> ⁴ -bis[(2-methoxyphenyl)methyl]butanediamide Solvents: Diethylene glycol; 24 h, rt		By: Ma, Xuerui; et al
Experimental Protocols		Advanced Synthesis & Catalysis (2022), 364(12), 2023-2031.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 98%



31-172-CAS-21455547

Steps: 1 Yield: 98%

Preparation, Characterization and Application of Copper Schiff base Complex Supported on MCM-41 as a Recyclable Catalyst for the Ullmann-type N-arylation Reaction

By: Niakan, Mahsa; et al

ChemistrySelect (2020), 5(1), 40-48.

1.1 Reagents: Potassium carbonate

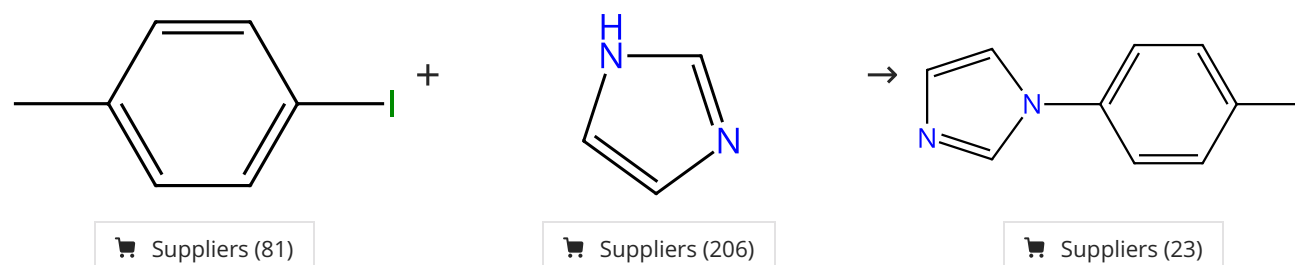
Catalysts: 2409651-49-4

Solvents: Dimethylformamide, Water; 3.5 h, 85 °C

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 98%



31-172-CAS-23296617

Steps: 1 Yield: 98%

L-Proline N-oxide dihydrazides as an efficient ligand for cross-coupling reactions of aryl iodides and bromides with amines and phenols

By: Ding, Zhiqiang; et al

Tetrahedron (2021), 79, 131826.

1.1 Reagents: Cesium carbonate

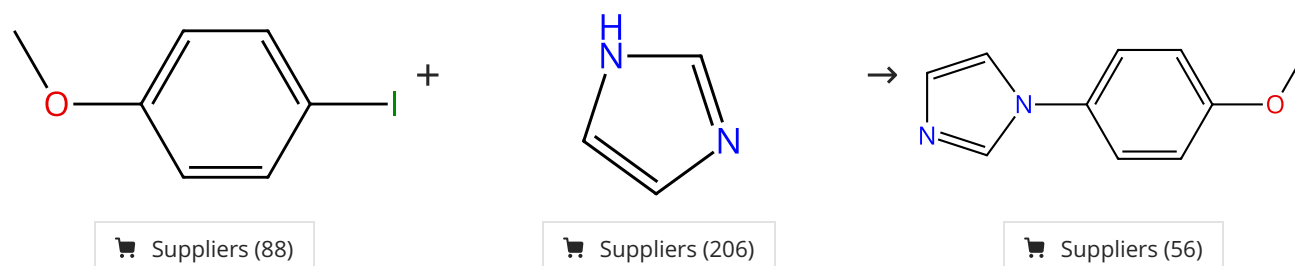
Catalysts: Cuprous iodide, 2588211-74-7

Solvents: Ethanol; 12 h, 80 °C

Experimental Protocols

Scheme 36 (2 Reactions)

Steps: 1 Yield: 98%



31-614-CAS-44931039

Steps: 1 Yield: 98%

MOF-253-Cu(OAc)₂ as a heterogeneous catalyst for N-arylation of heterocycles

By: Cao, Ningtao; et al

Journal of Organometallic Chemistry (2025), 1033, 123641.

1.1 Reagents: Sodium hydroxide

Catalysts: [[2,2'-Bipyridine]-5,5'-dicarboxylato(2-)-κO⁵] hydroxylaluminum (post-treated with Cu(OAc)₂)

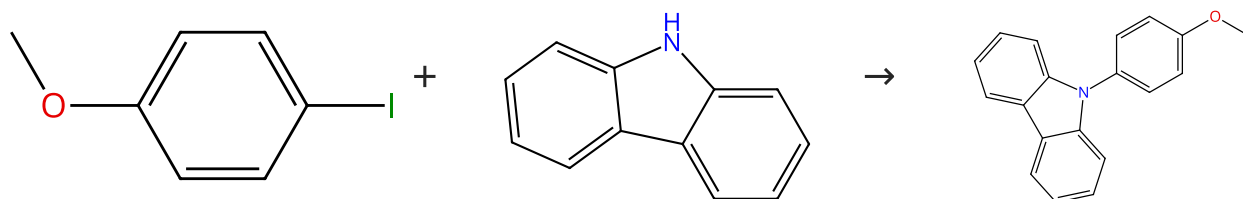
Solvents: Dimethyl sulfoxide; 12 h, 120 °C

Experimental Protocols

31-172-CAS-21750793	Steps: 1 Yield: 98%	Facile synthesis of hydrochar supported copper nanocatalyst for Ullmann C-N coupling reaction in water
1.1 Reagents: Potassium carbonate Catalysts: Copper Solvents: Water; rt → 100 °C; 24 h, 100 °C		By: Ge, Xin; et al Molecular Catalysis (2020), 484, 110726.
Experimental Protocols		

Scheme 37 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (88)

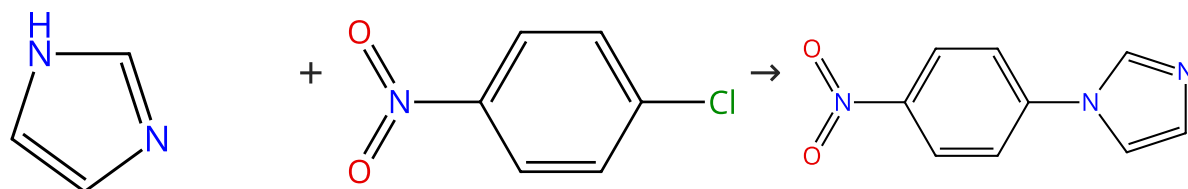
Suppliers (117)

Suppliers (17)

31-172-CAS-23028517	Steps: 1 Yield: 98%	The remarkable selectivity of the 2-arylquinoline-based acyl hydrazones toward copper salts: exploration of their catalytic applications in the copper catalyzed N-arylation of indole derivatives and C1-alkynylation of tetrahydroisoquinolines via the A ³ reaction
1.1 Reagents: Potassium carbonate Catalysts: Cuprous iodide, 2568377-50-2 Solvents: Dimethyl sulfoxide; 24 h, 110 °C		By: Echeverry-Gonzalez, Carlos A.; et al New Journal of Chemistry (2021), 45(1), 243-250.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (206)

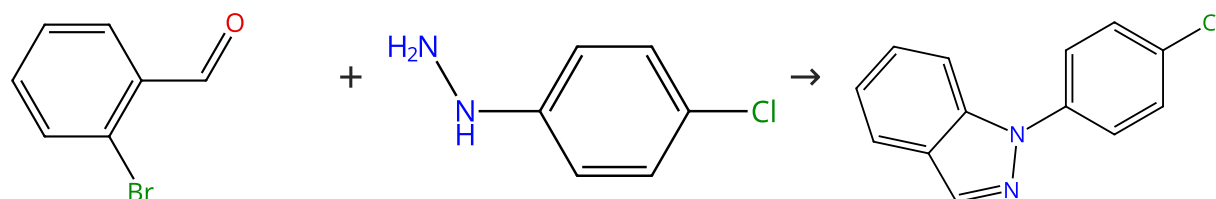
Suppliers (65)

Suppliers (80)

31-172-CAS-22720801	Steps: 1 Yield: 98%	N-Heterocyclic Carbene Copper(I) Complex Catalyzed Coupling of (Hetero)aryl Chlorides and Nitrogen Heterocycles: Highly Efficient Catalytic System
1.1 Reagents: Cesium carbonate Catalysts: [8-[2,6-Bis(1-methylethyl)phenyl]imidazo[1,5-a][1,8]naphthyridin-9-ylidene-κC ⁹ ,κN ¹]chlorocopper Solvents: Dimethylformamide; 12 h, 120 °C		By: Zhang, Mengyao; et al Chinese Journal of Chemistry (2020), 38(11), 1252-1256.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (89)

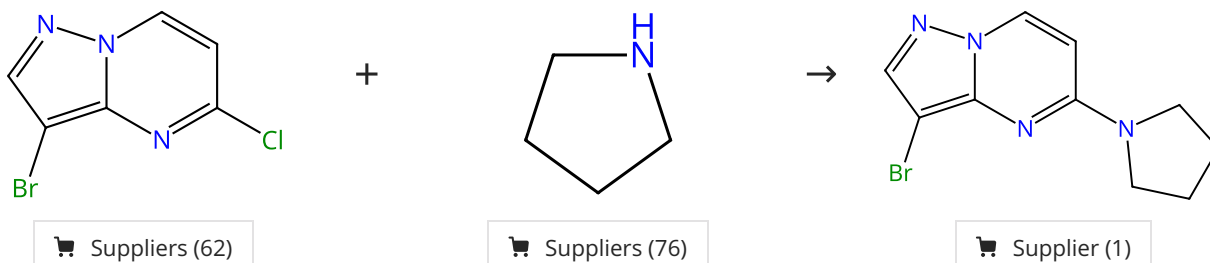
Suppliers (65)

Suppliers (2)

31-614-CAS-34659788	Steps: 1 Yield: 97%	Bottom-Up De Novo Synthesis of Porous Organic Polymers with Enone Functionalities as Supports for Pd and Cu Nanoparticles for Catalytic Tandem Synthesis
1.1 Catalysts: 2832920-74-6 (support palladium and copper) Solvents: Dimethylformamide; 10 min, rt		By: Yadav, Chetna; et al
1.2 Reagents: Cesium carbonate; 2 h, 120 °C		ACS Applied Nano Materials (2022), 5(10), 14296-14310.
Experimental Protocols		

Scheme 40 (1 Reaction)

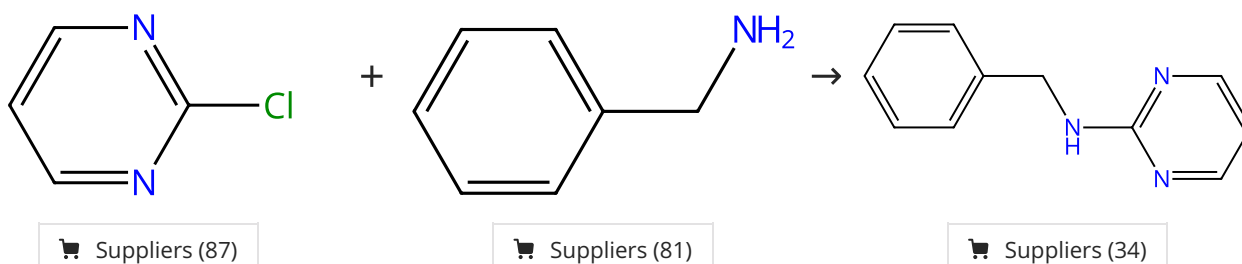
Steps: 1 Yield: 97%



31-614-CAS-44174347	Steps: 1 Yield: 97%	An Efficient Synthesis of 3,5-Bis-Aminated Pyrazolo[1,5-a]Pyrimidines: Microwave-Assisted Copper Catalyzed C-3 Amination of 5-Amino-3-Bromo-Substituted Precursors
1.1 Reagents: Cesium fluoride Catalysts: Benzyltriethylammonium chloride Solvents: Dimethyl sulfoxide; 8 h, 100 °C		By: Iorkula, Terungwa H.; et al
Experimental Protocols		Molecules (2025), 30(3), 458.

Scheme 41 (1 Reaction)

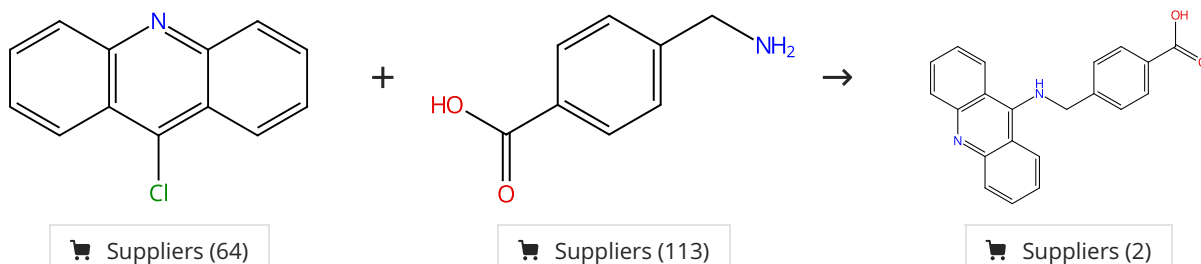
Steps: 1 Yield: 97%



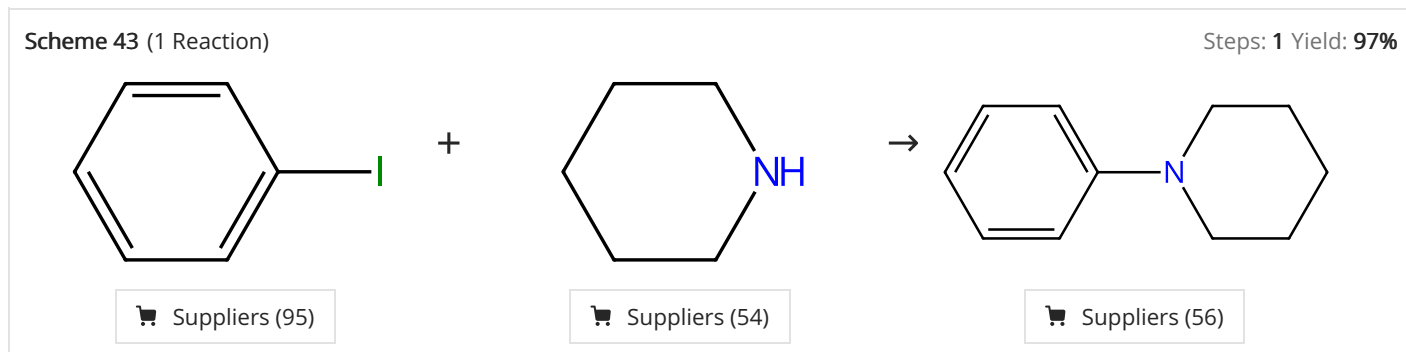
31-614-CAS-39150067	Steps: 1 Yield: 97%	Cu-catalyzed, ligand free synthesis of arylaminopyrimidines in aqueous solution and in-silico investigation of their druglike properties
1.1 Reagents: Potassium carbonate Catalysts: Cupric nitrate Solvents: Water; 24 h, rt → 80 °C		By: Bhuyan, Amar Jyoti; et al
Experimental Protocols		Sustainable Chemistry and Pharmacy (2024), 37, 101354.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 97%



31-172-CAS-22688699	Steps: 1 Yield: 97%	Synthesis and Anticancer Research of N-(2-aminophenyl) benzamide Acridine Derivatives as Dual Topoisomerase I and Isoform-Selective HDAC Inhibitors By: Zhang, Bin; et al ChemistrySelect (2020), 5(27), 8311-8318.
1.1 Solvents: Glycol monoethyl ether; 90 - 100 °C		
Experimental Protocols		



31-614-CAS-34769813	Steps: 1 Yield: 97%	Magnetic MCR-Functionalized Graphene Oxide Complexed with Copper Nano-Particles: An Efficient and Recyclable Nanocatalyst for Ullmann C-N Coupling Reaction By: Bagheri, Hashem; et al Polycyclic Aromatic Compounds (2023), 43(8), 7580-7596.
1.1 Reagents: Potassium carbonate Catalysts: Copper Solvents: Tetrahydrofuran		
Experimental Protocols		